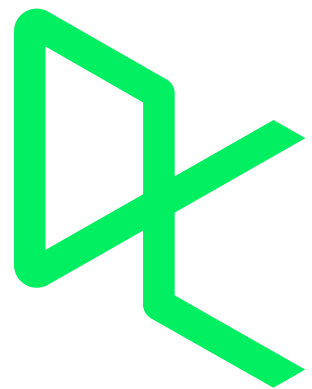


WHAT IS THE OpenAI API?





James Chapman

*Curriculum Manager,
DataCamp*



Course goals

- 1 Use AI models through the **OpenAI API**

Course goals

- ➊ Use AI models through the **OpenAI API**
- ➋ Solve real-world tasks

Course goals

- ➊ Use AI models through the **OpenAI API**
- ➋ Solve real-world tasks
- ➌ Use **Python** code

Expected knowledge

- Subsetting lists and dictionaries



Expected knowledge

- Subsetting lists and dictionaries
- Control flow

`if`

`elif`

`else`



Expected knowledge



- Subsetting lists and dictionaries
- Control flow

`if`

`elif`

`else`

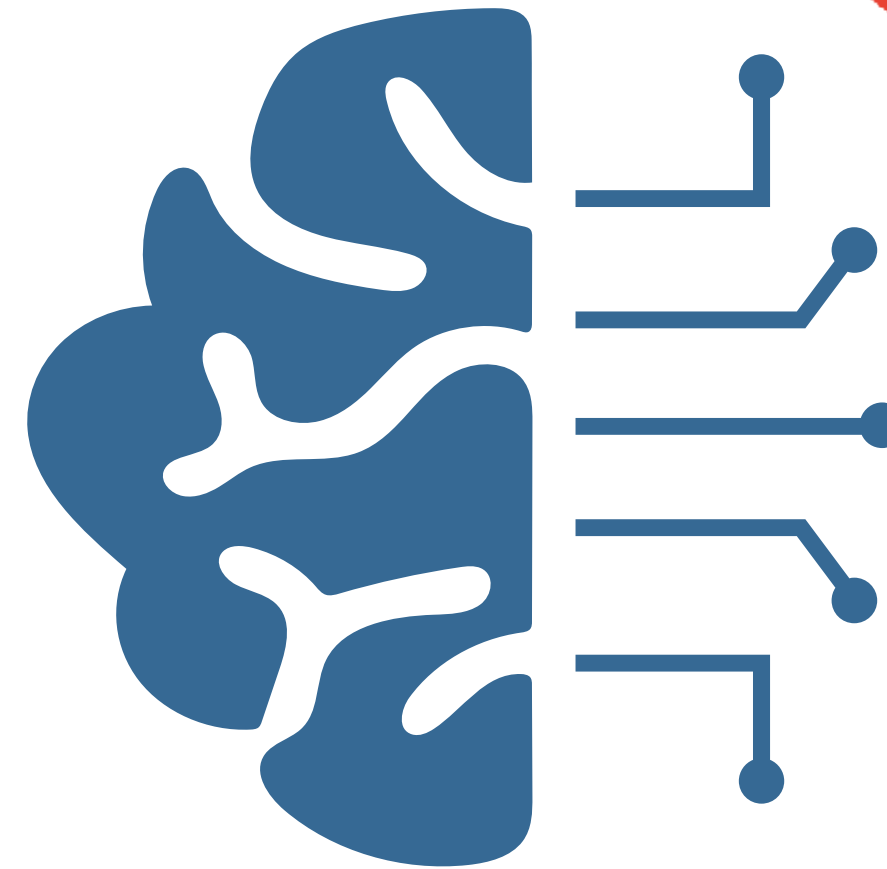
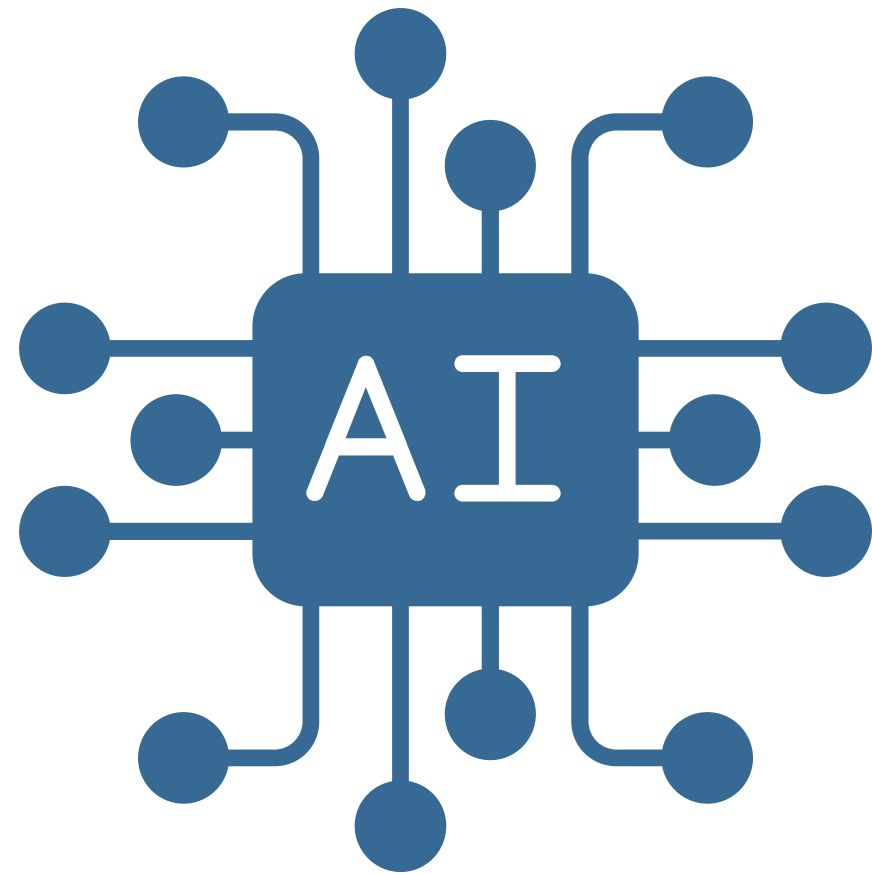
- Loops

`for`

`while`



Not expected



Machine Learning

Let's dive in!

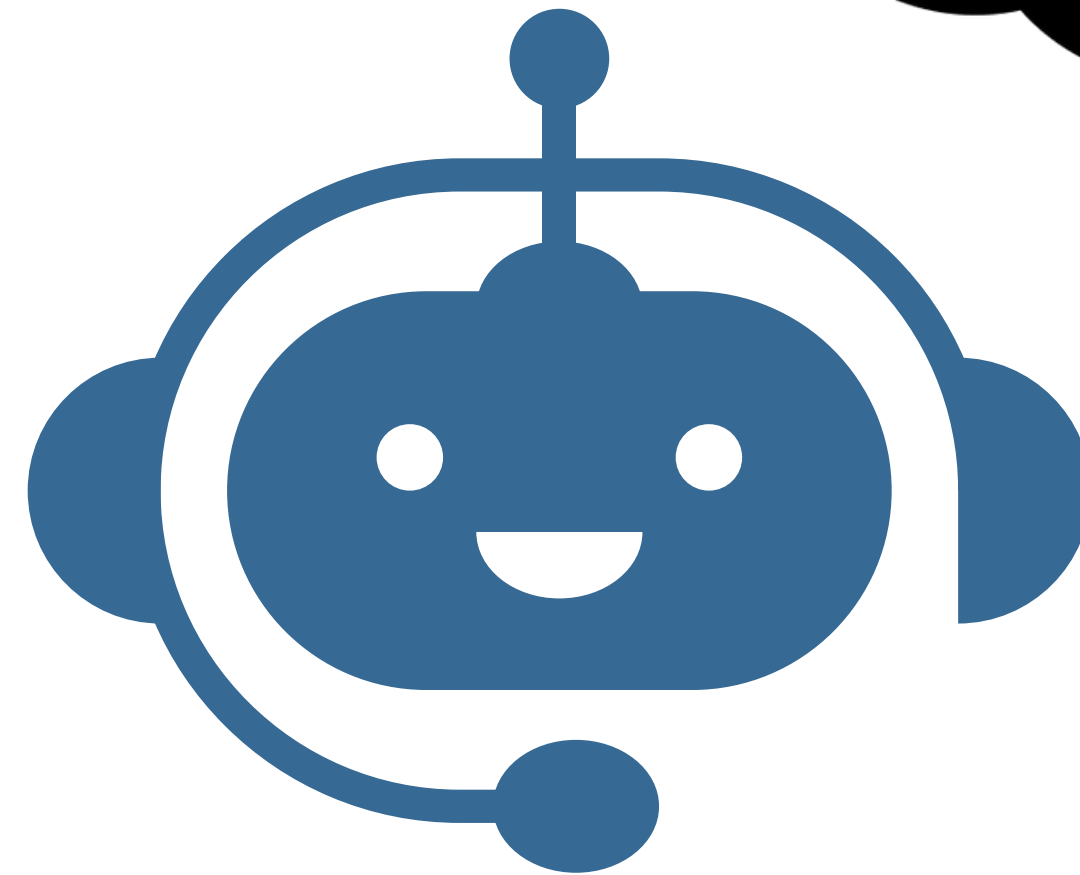
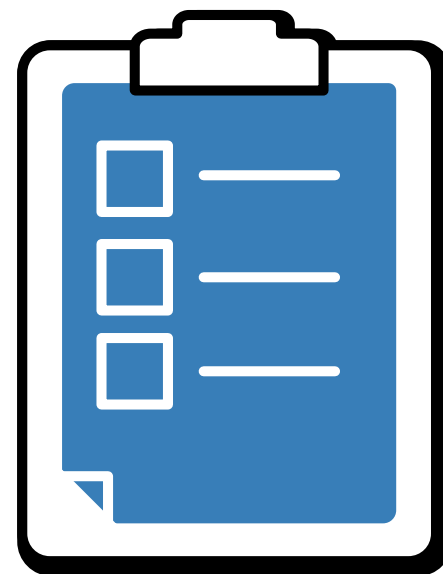
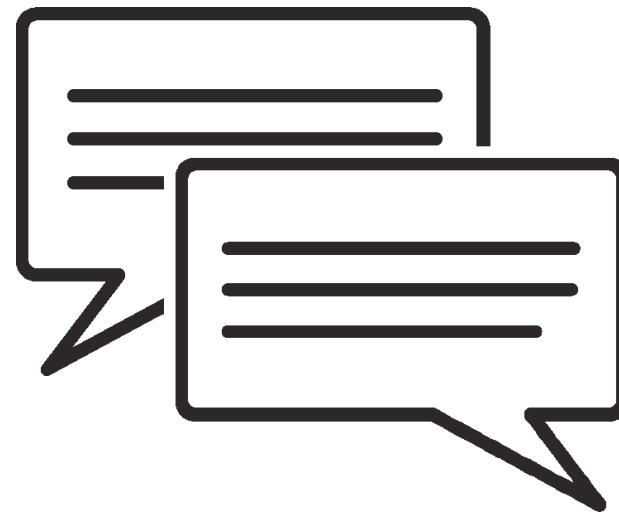
OpenAI

- Research and develop AI systems



OpenAI

- AI-powered chatbot



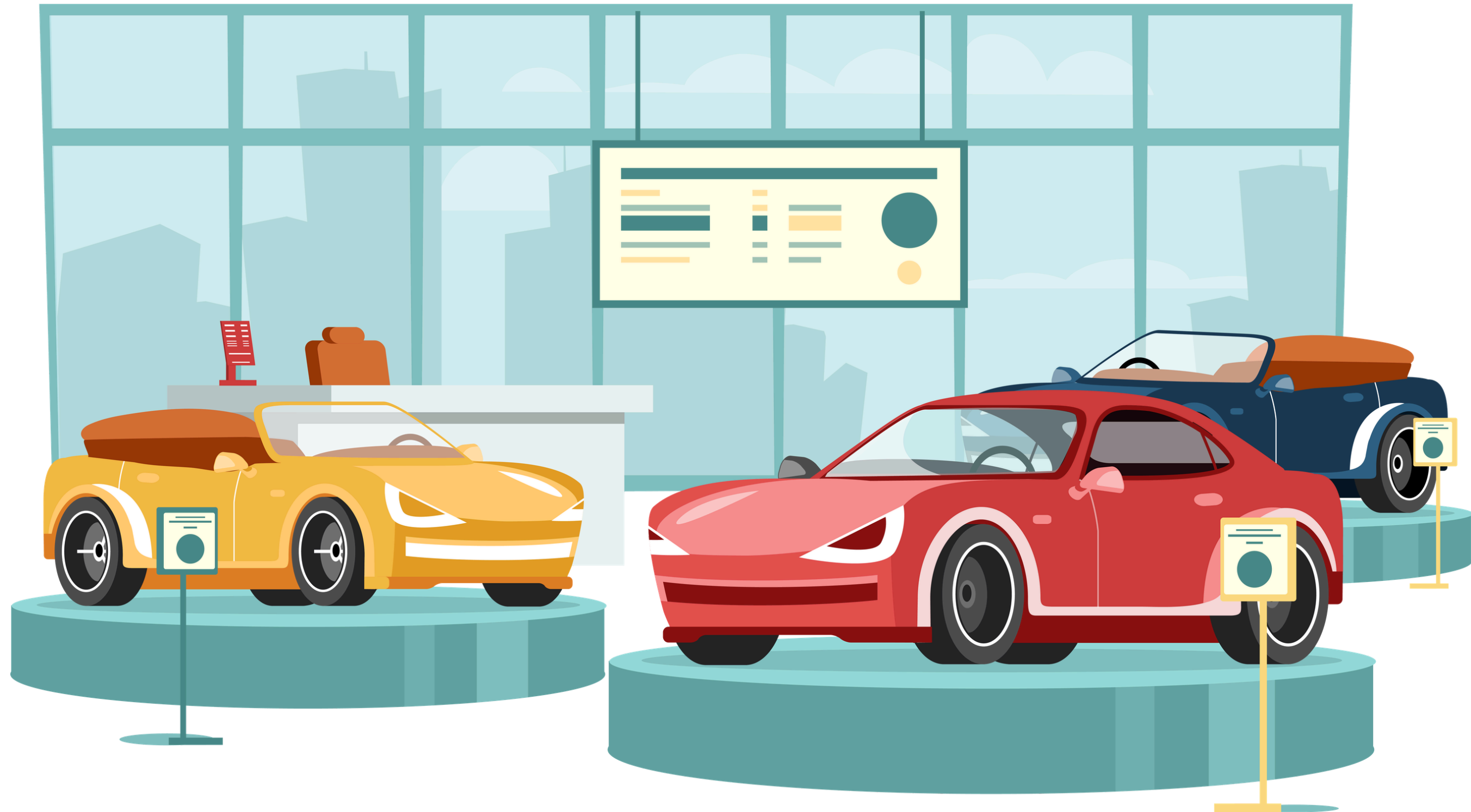
ChatGPT

OpenAI API

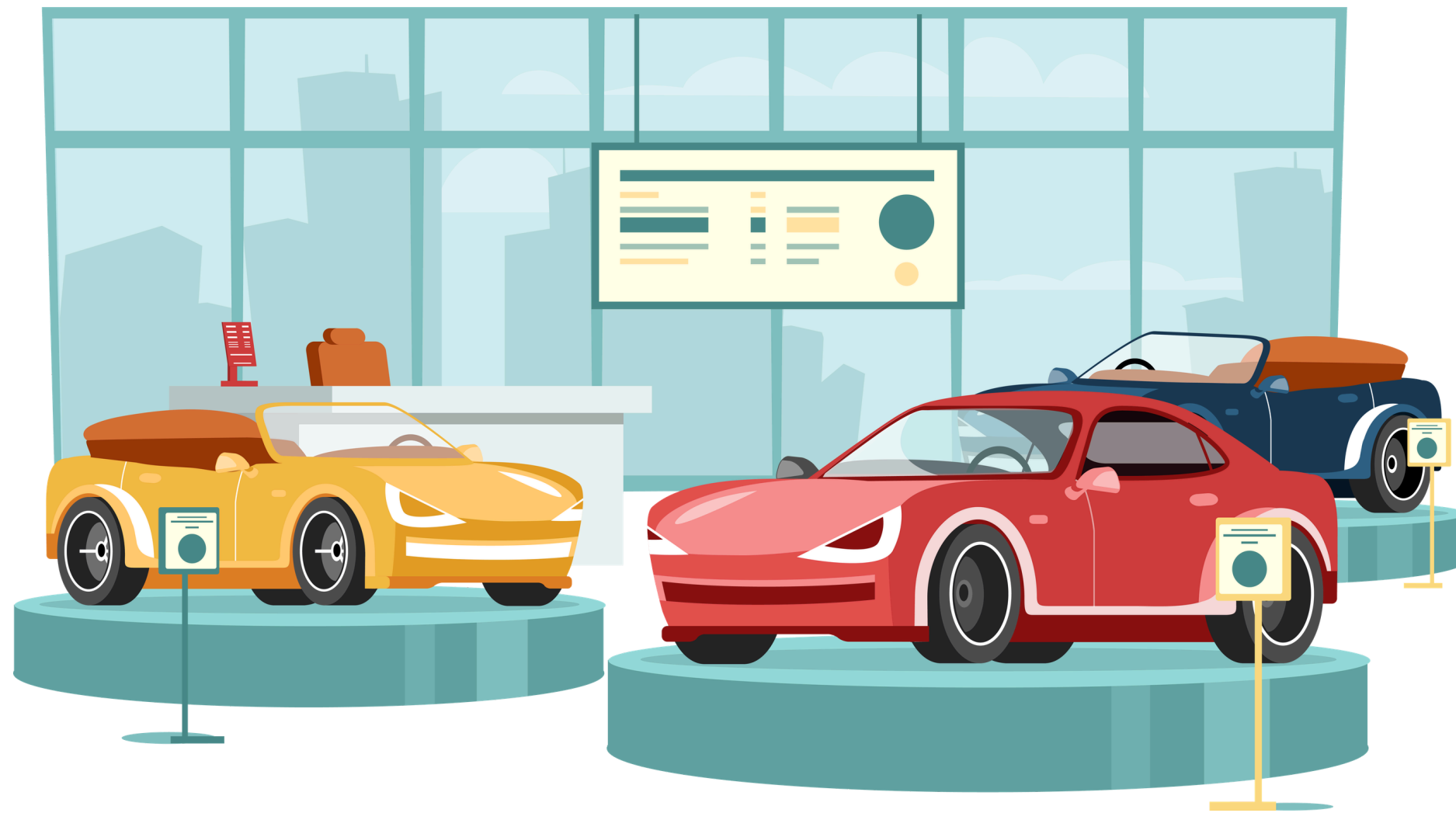


- Access and customize OpenAI models

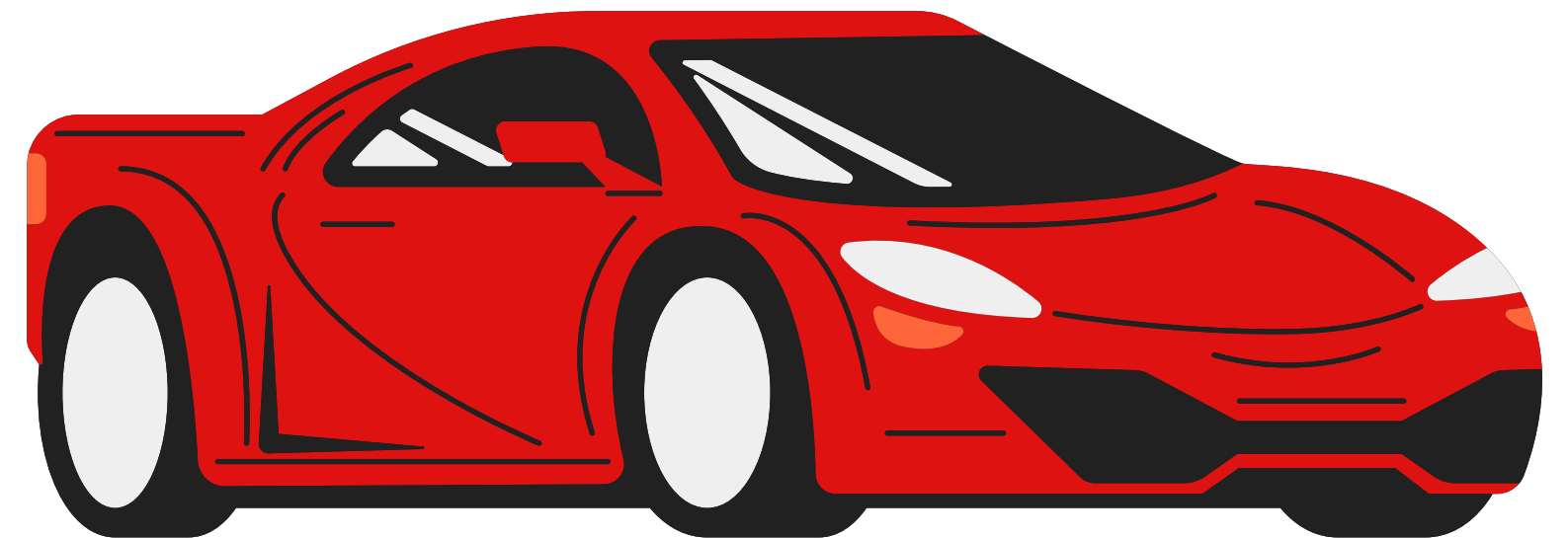
OpenAI



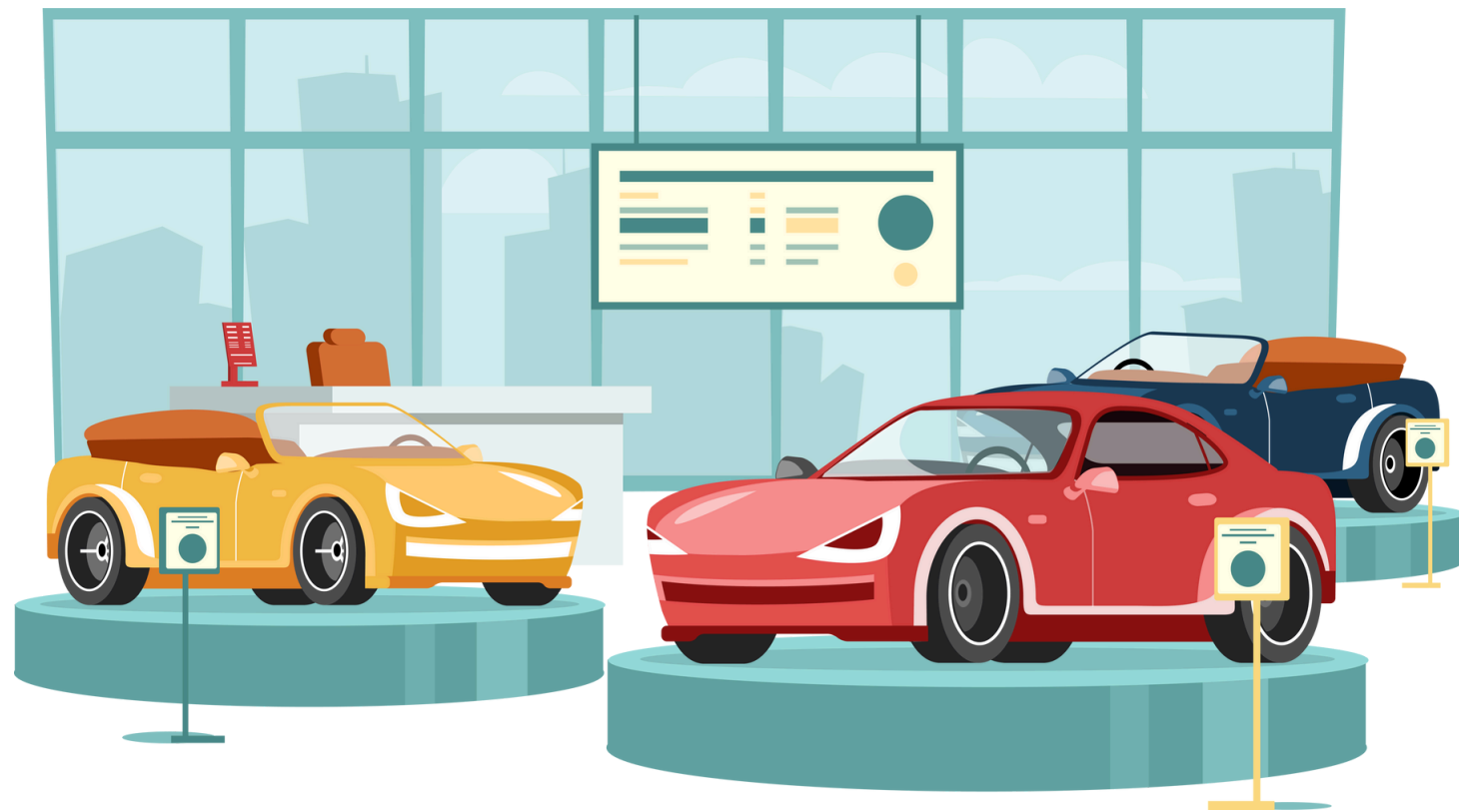
OpenAI



ChatGPT



OpenAI



ChatGPT

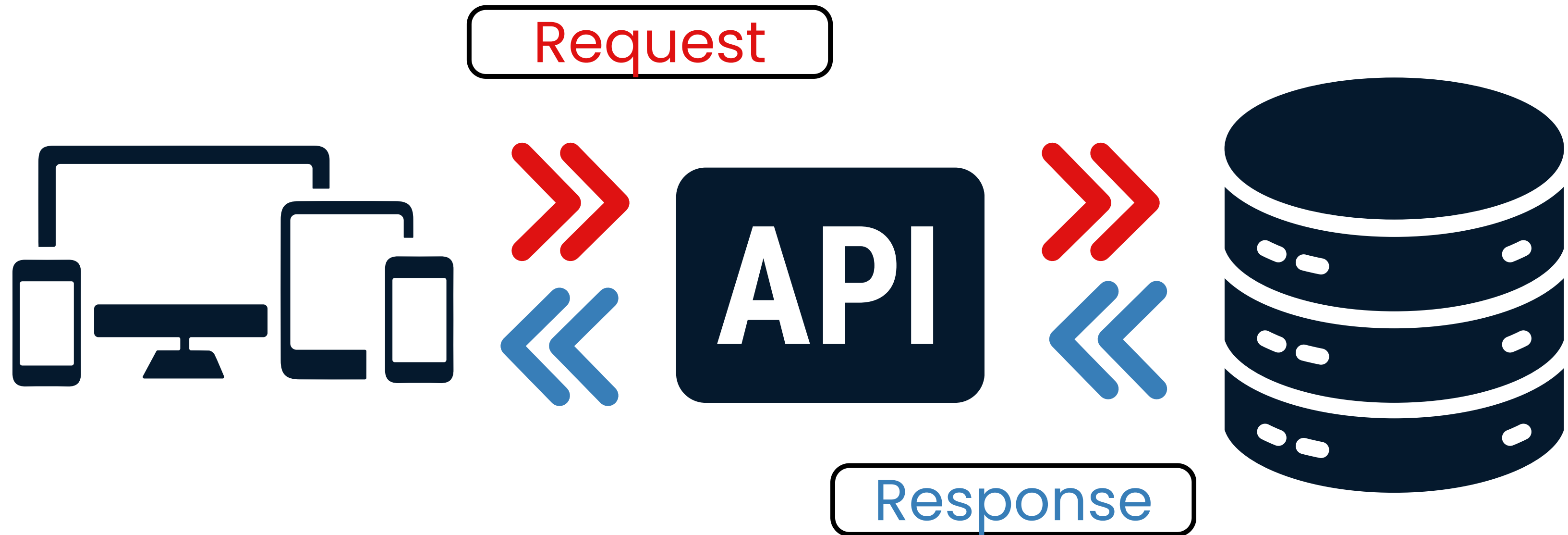


OpenAI API



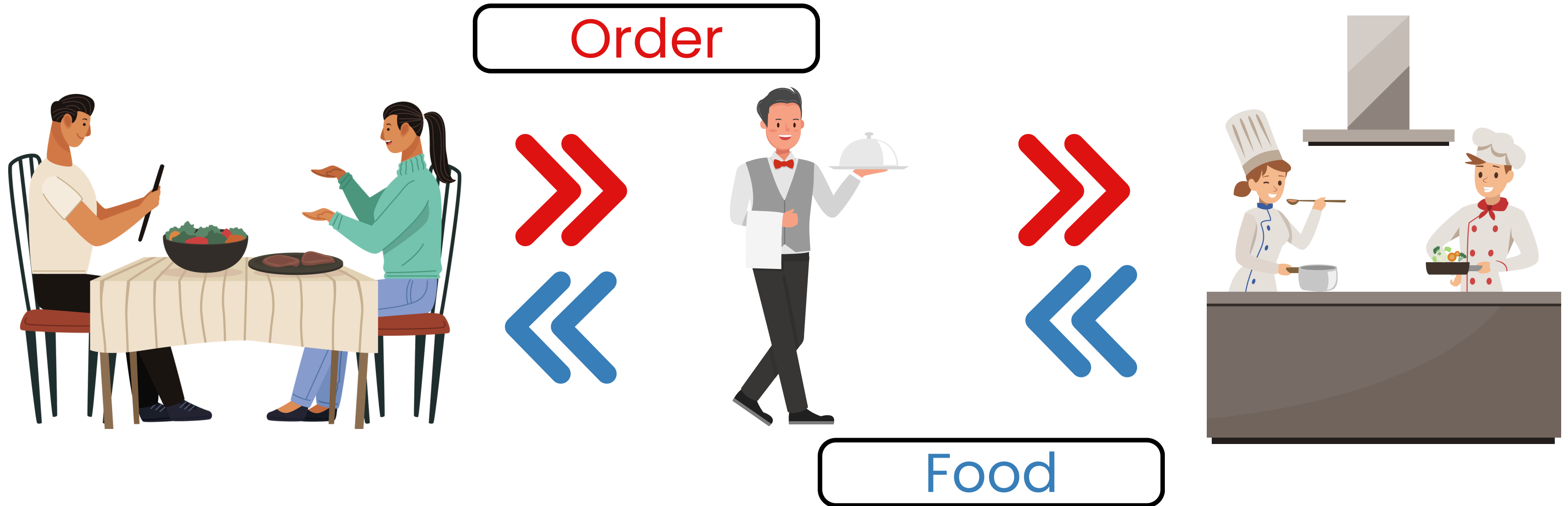
What is an API?

- **A**pplication **P**rogramming **I**nterface



What is an API?

- **Application Programming Interface**

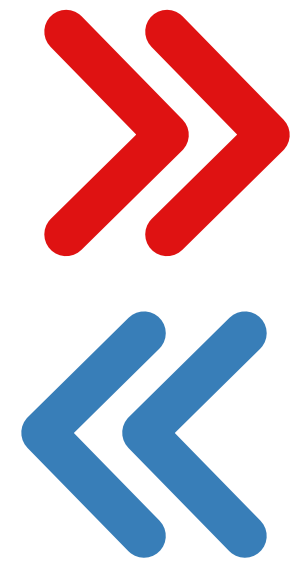


What is an API?

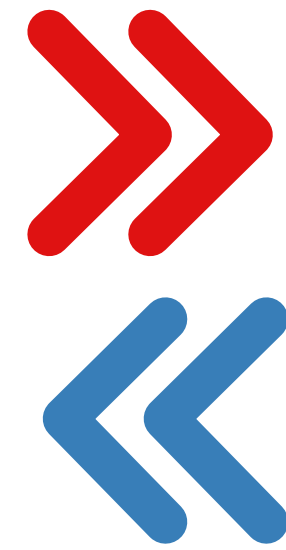
- **A**pplication **P**rogramming **I**nterface



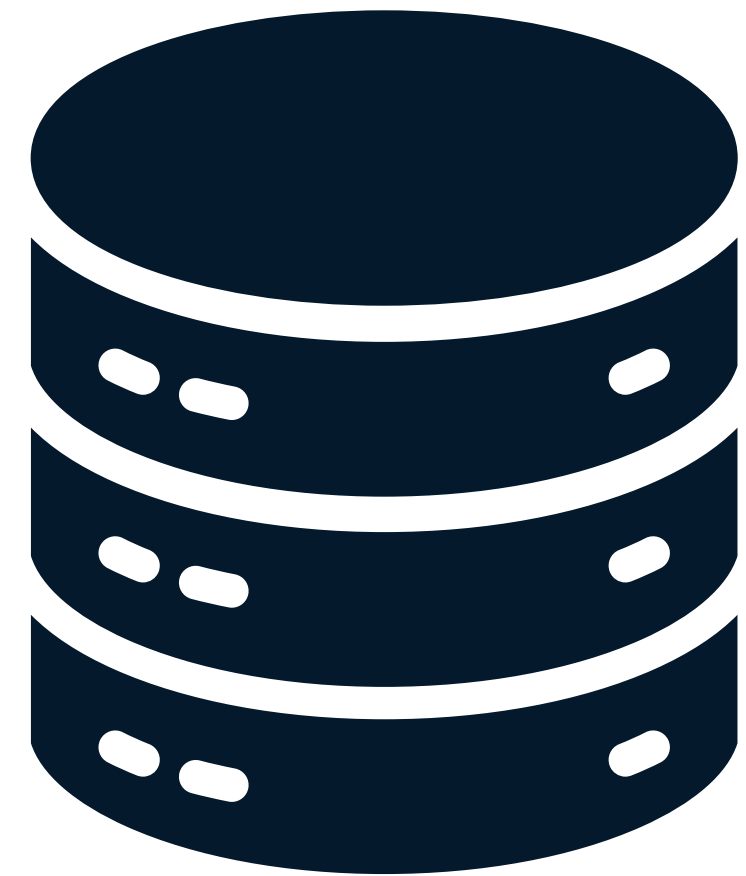
Location



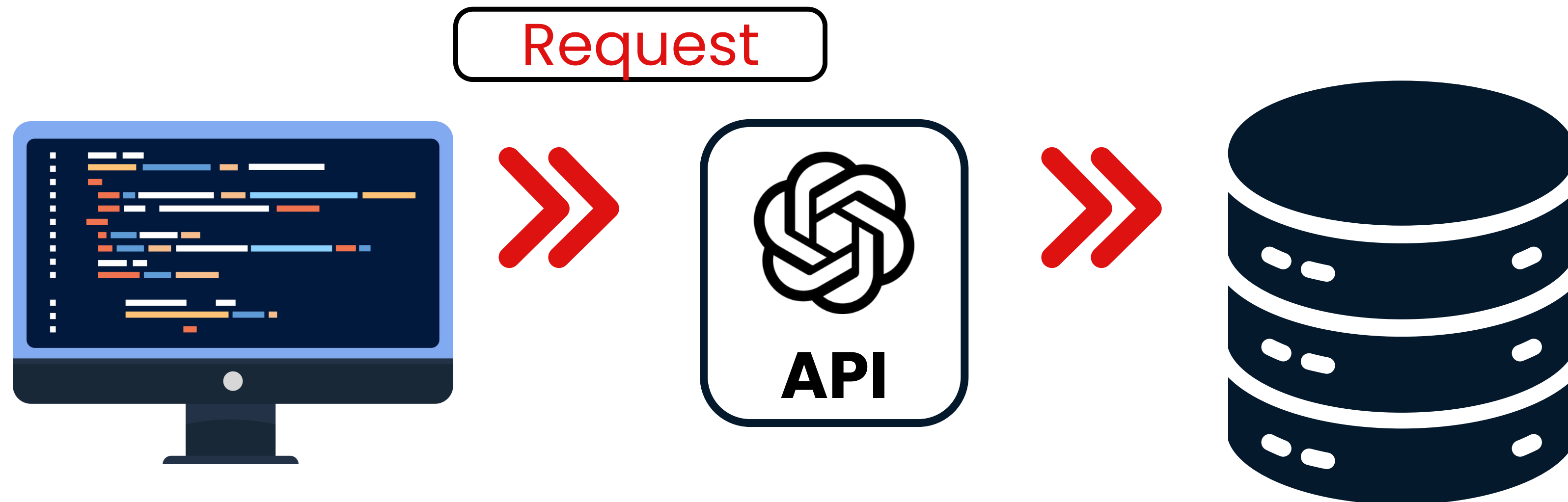
API



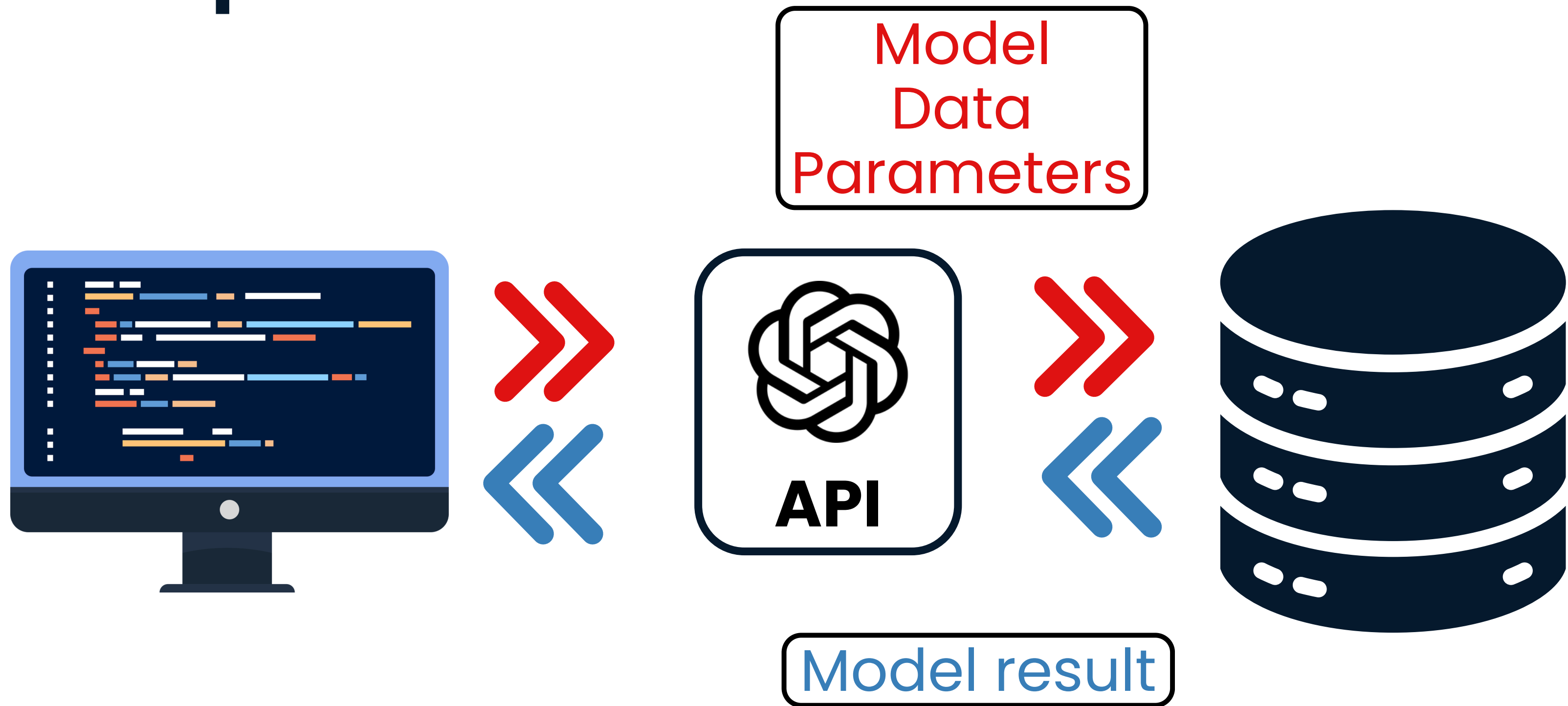
Weather



The OpenAI API



The OpenAI API



Web interface vs. APIs

ChatGPT

Write Python code for drawing a scatter plot.

Here's a simple Python code snippet using `matplotlib` to draw a scatter plot:

```
python Copy

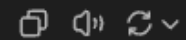
import matplotlib.pyplot as plt

# Sample data
x = [1, 2, 3, 4, 5]
y = [2, 3, 5, 7, 11]

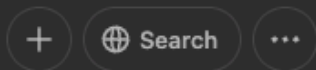
# Create scatter plot
plt.scatter(x, y)

# Add labels
plt.xlabel('X-axis')
plt.ylabel('Y-axis')

# Title
```



Message ChatGPT



- Minimal setup

Web interface vs. APIs

ChatGPT

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📄 🔊 ↺

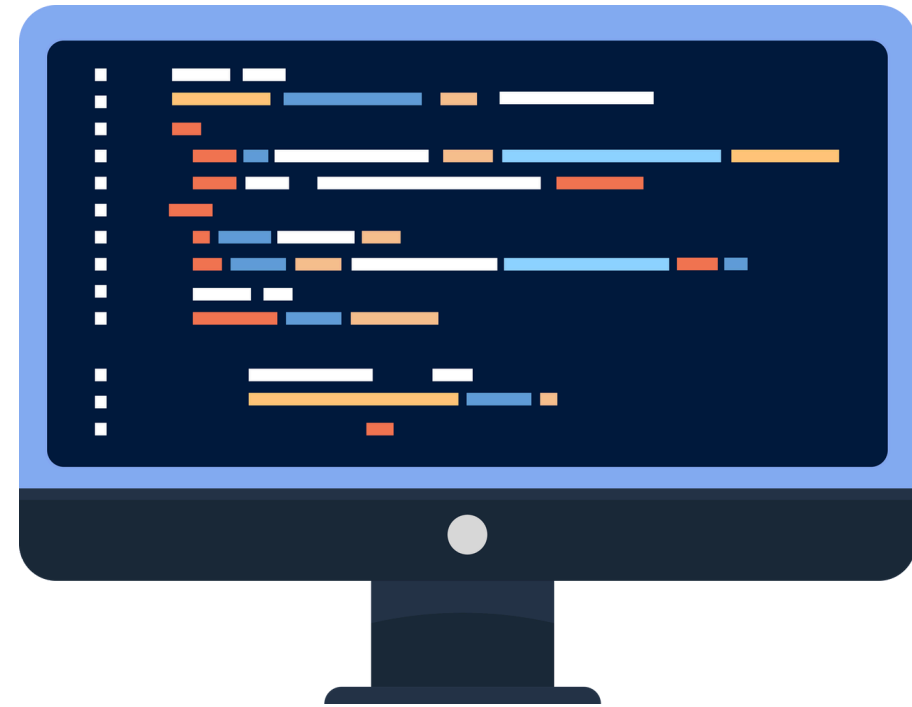
Message ChatGPT

+ 🌐 Search ...



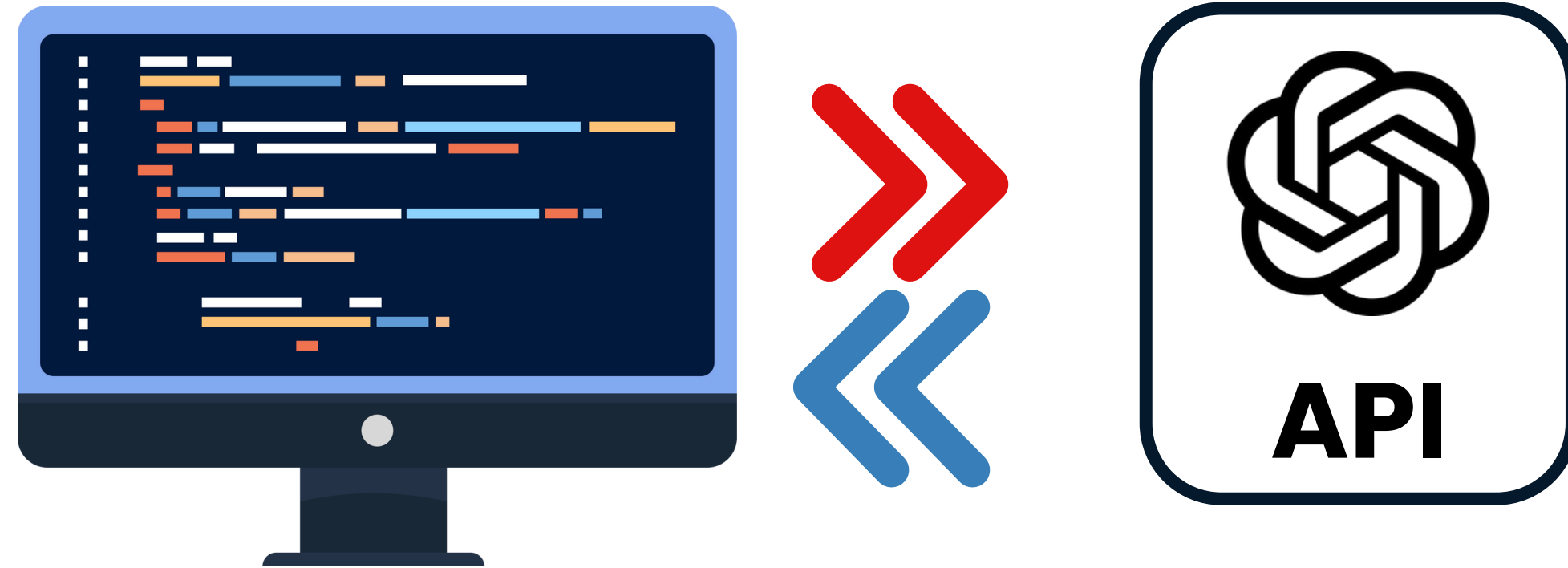
- Minimal setup
- Streamline workflows

Web interface vs. APIs



- Integrate AI into products

Web interface vs. APIs



- Integrate AI into products
- Work with models programmatically

OpenAI API

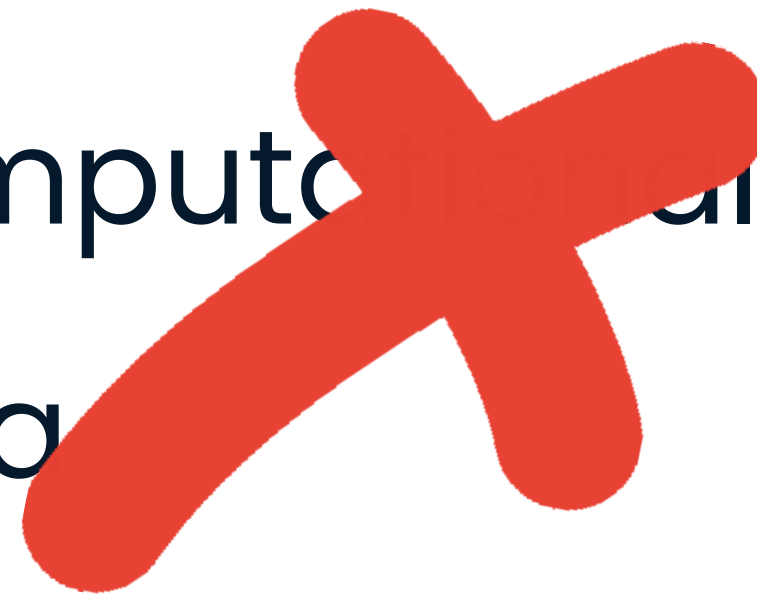


NEW

OpenAI API



- Computational resources
- Data



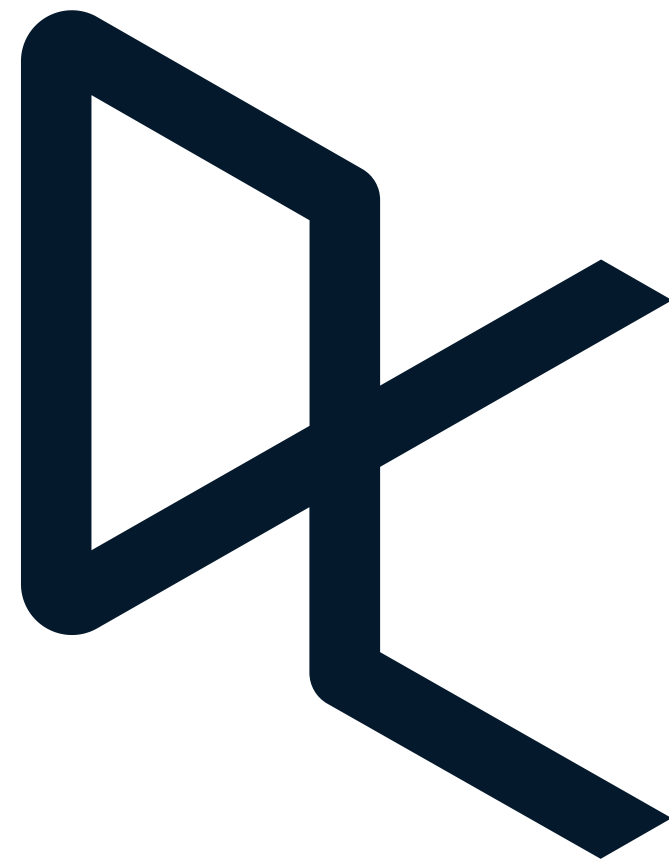
Building AI applications



- AI-powered IDE



LET'S PRACTICE!

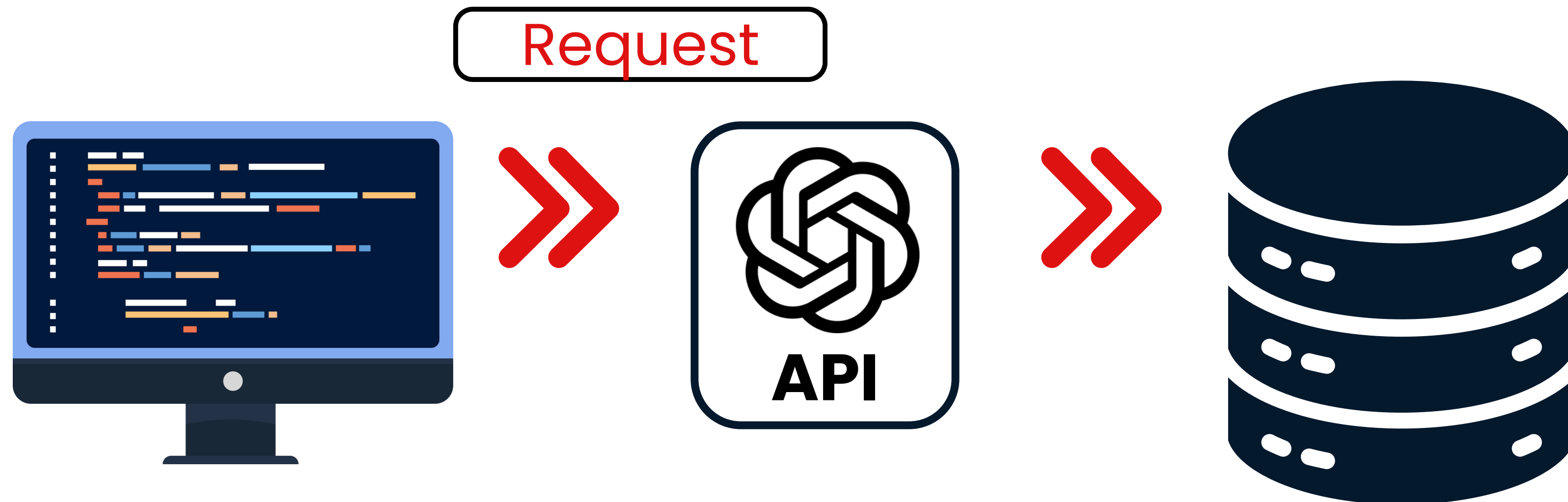


MAKING REQUESTS TO THE

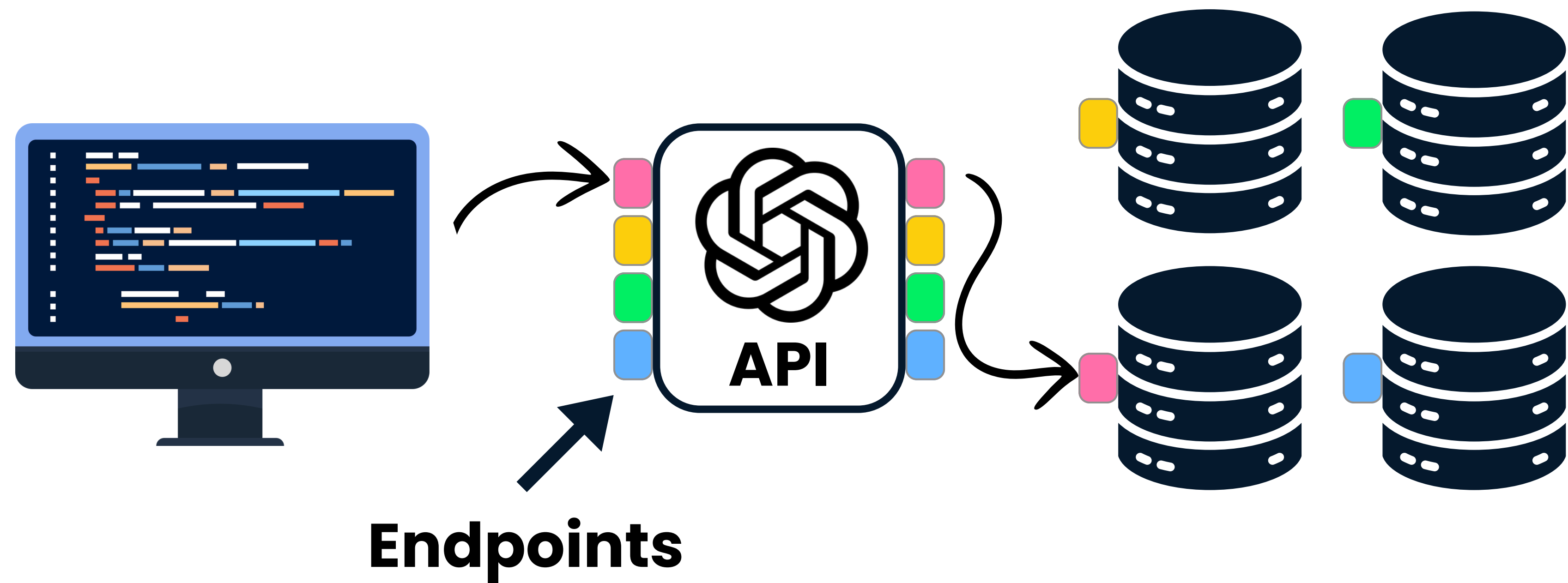
OpenAI API

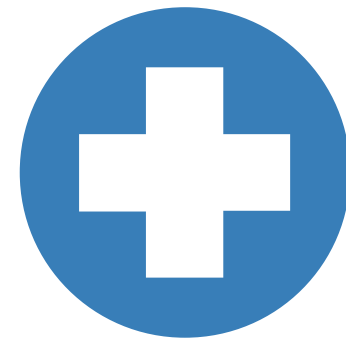
 datacamp

The OpenAI API

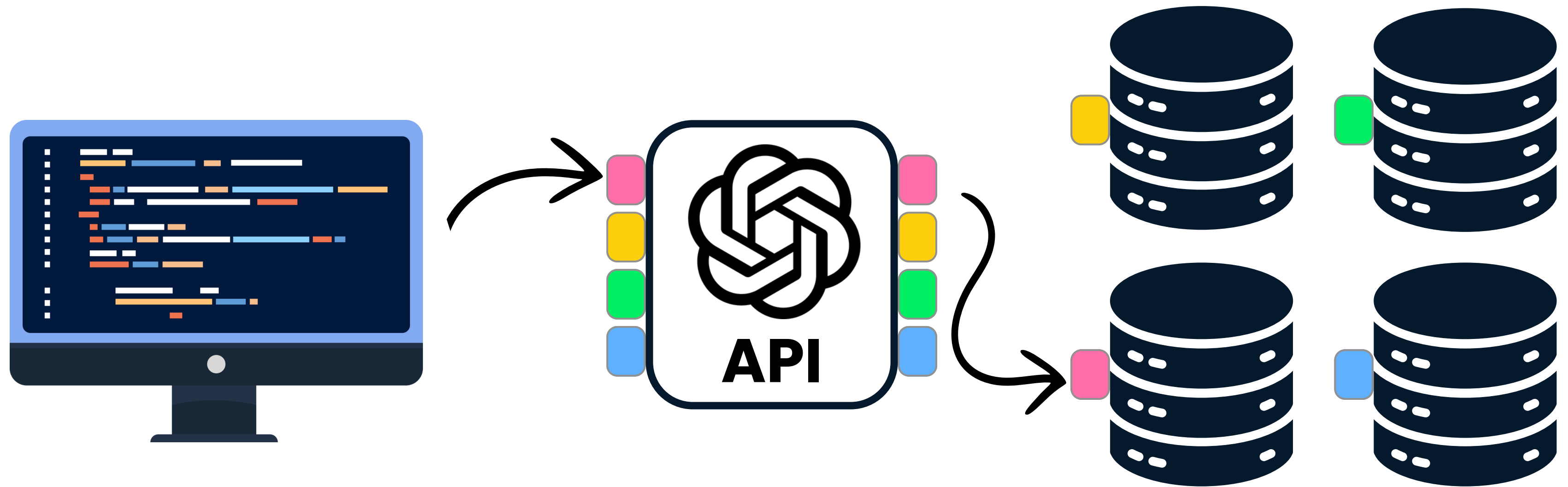


API Endpoints

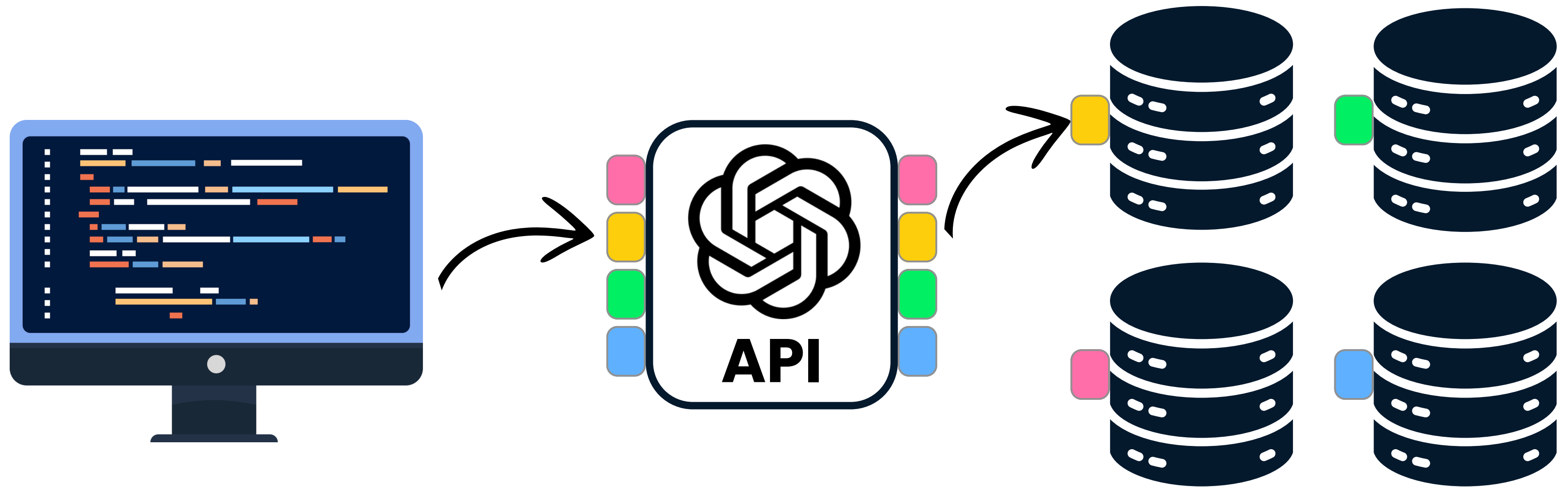




API Endpoints



API Endpoints



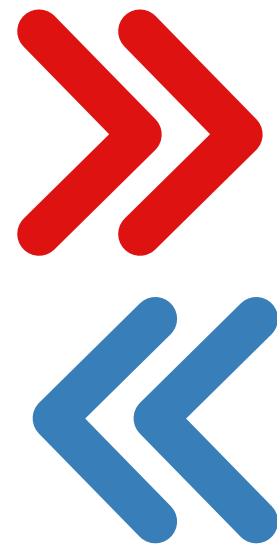
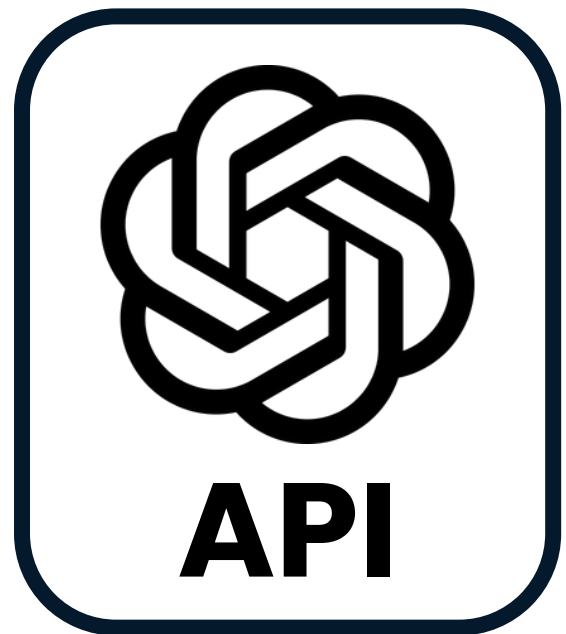
Authentication

- Unique key

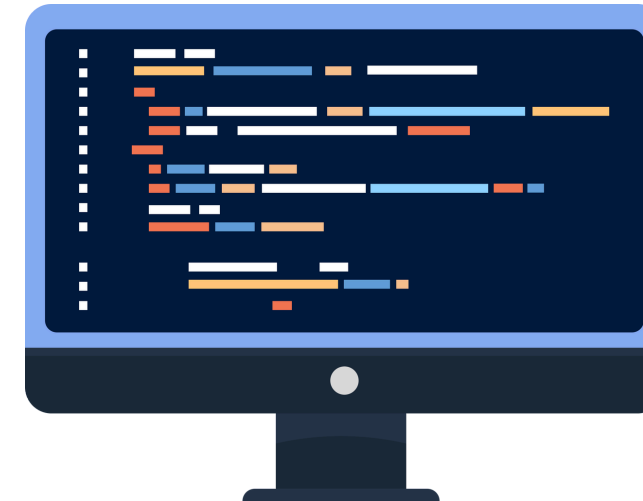


API usage costs

- OpenAI API



Course exercises



Exercise

Categorizing companies

In this exercise, you'll use a Chat Completions model to categorize different companies. At first, you won't specify the categories to see how the model categorizes them. Then, you'll specify the categories in the prompt to ensure they are categorized in a desirable and predictable way.

Instructions100 XP

- Create a request to the Chat Completions endpoint to categorize the following companies: Apple, Microsoft, Saudi Aramco, Alphabet, Amazon, Berkshire Hathaway, NVIDIA, Meta, Tesla, and LVMH; run the code to see the response.
- Alter the prompt to specify the four categories that the companies should be classified into: Tech, Energy, Luxury Goods, or Investment, and re-run the code.

Take Hint (-30 XP)

script.pyLight mode

```
1 client = OpenAI(api_key= <OPENAI_API_TOKEN> )
2
3 # Define a prompt for the categorization
4 prompt = "Categorize these companies by sector into Tech, Energy, Luxury
5 Goods, or Investment: Apple, Microsoft, Saudi Aramco, Alphabet, Amazon,
6 Berkshire Hathaway, NVIDIA, Meta, Tesla, LVMH"
7
8 # Create a request to the Chat Completions endpoint
9 response = client.chat.completions.create(
10     model="gpt-4o-mini",
11     messages=[{"role": "user", "content": prompt}],
12     max_tokens=100,
13     temperature=0.5
14 )
15 print(response.choices[0].message.content)
```

contentlocal

IPython ShellSlides

In [1]:



Course exercises

Exercise

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script.py

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```

content local

IPython Shell Slides

In [1]:

A blue gear icon with the letters 'API' inside, crossed out with a large red 'X'.

Creating an API key

- 1 Create an OpenAI account
- 2 Go to the API keys page
- 3 Create a new secret key

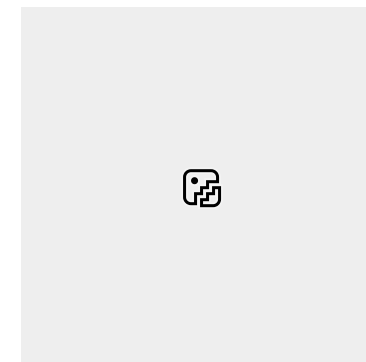
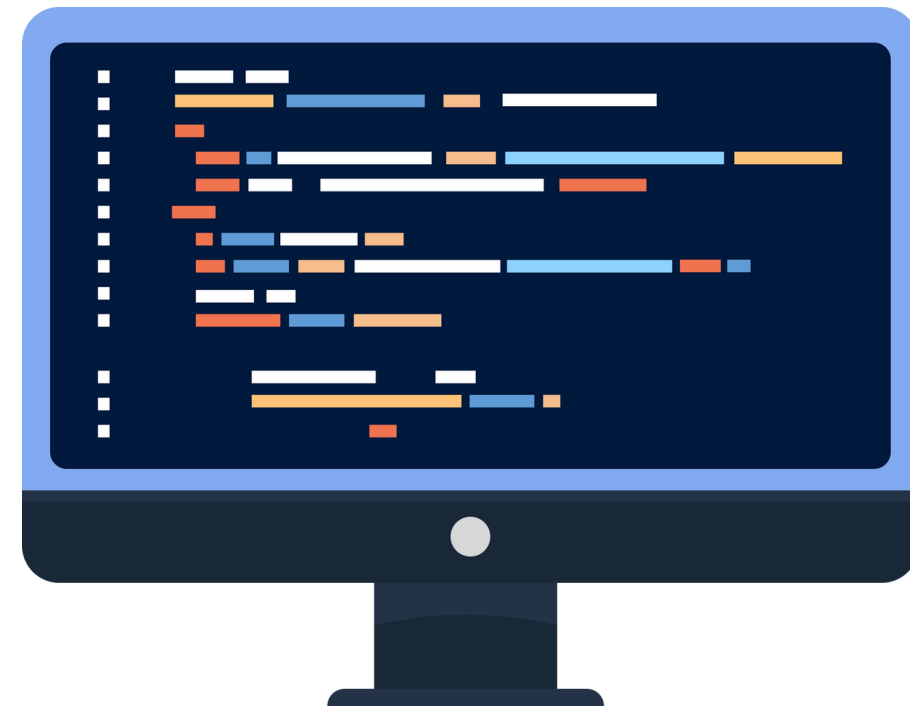


NAME	KEY	CREATED	LAST USED
Secret key	sk-...Vpp5	5 Feb 2023	25 Apr 2023

+ Create new secret key



Making a request



Making a request

```
from openai import OpenAI  
  
client = OpenAI(api_key="ENTER YOUR KEY HERE")
```

- **Client** – configures environment for communicating with API

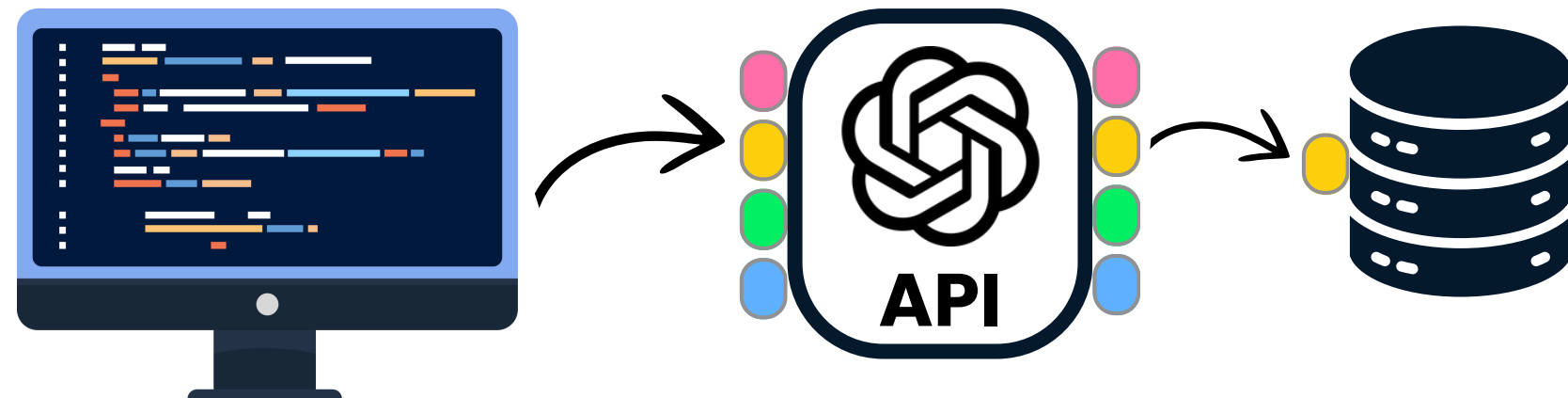
Making a request

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from openai import OpenAI  
  
client = OpenAI(api_key="ENTER YOUR KEY HERE")
```



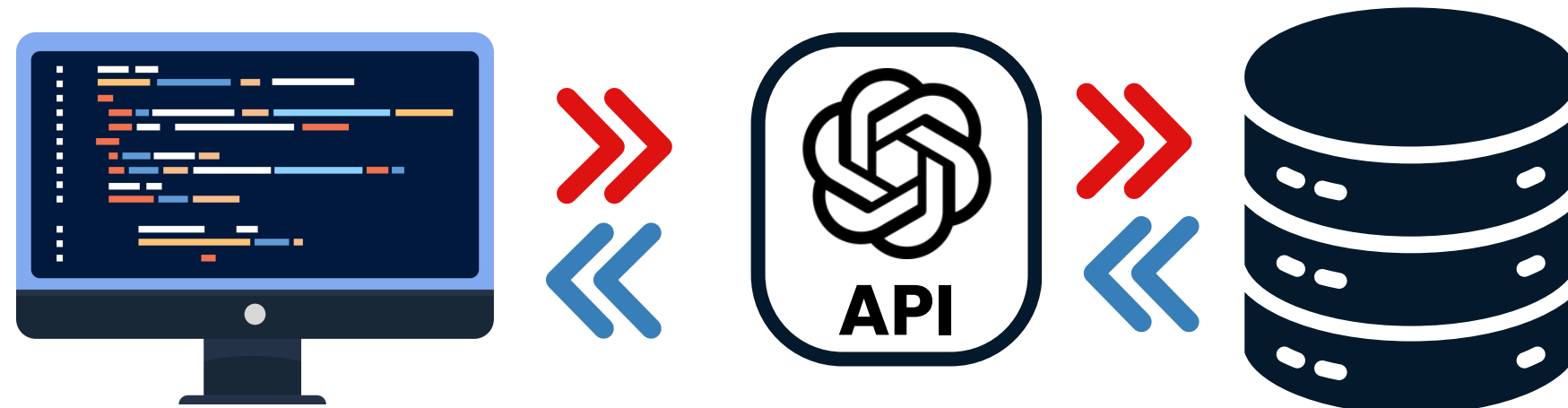
Making a request

```
from openai import OpenAI  
  
client = OpenAI(api_key="ENTER YOUR KEY HERE")  
  
response = client.chat.completions.create(  
  
)
```



Making a request

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from openai import OpenAI  
  
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)
```



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- List of dictionaries
- Content from **user** role prompts the model

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)

print(response)
```

The response

```
ChatCompletion(id='chatcmpl-AEcQbQekIzxcxVAKYVAjgUAXokgrl', choices=[Choice(finish_reason='length', index=0, logprobs=None, message=ChatCompletionMessage(content='The OpenAI API is a cloud-based service provided by OpenAI that allows developers to integrate advanced AI models into their applications.', refusal=None, role='assistant', function_call=None, tool_calls=None))], created=1728047673, model='gpt-4o-mini-2024-07-18', object='chat.completion', service_tier=None, system_fingerprint='fp_f85bea6784', usage=CompletionUsage(completion_tokens=30, prompt_tokens=14, total_tokens=44, prompt_tokens_details {'cached_tokens': 0}, completion_tokens_details={'reasoning_tokens': 0}))
```

The response

```
ChatCompletion(id='chatcmpl-AEcQbQekIzxcxVAKYVAjgUAXokgrl',
               choices=[Choice(finish_reason='length', index=0, logprobs=None,
                               message=ChatCompletionMessage(content='The OpenAI API is a cloud-
                               based service provided by OpenAI that allows developers to
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                               refusal=None, role='assistant', function_call=None,
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               created=1728047673,
               model='gpt-4o-mini-2024-07-18',
               object='chat.completion', service_tier=None, system_fingerprint='fp_f85bea6784',
               usage=CompletionUsage(completion_tokens=25, prompt_tokens=14, total_tokens=39,
                                     prompt_tokens_details={'cached_tokens': 0},
                                     completion_tokens_details={'reasoning_tokens': 0}))
```

The response

```
ChatCompletion(id='chatcmpl-AEcQbQekIzxcxVAKYVAjgUAXokgrl',
               choices=[Choice(finish_reason='length', index=0, logprobs=None,
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                                     prompt_tokens_details={'cached_tokens': 0},
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```

Interpreting the response

```
print(response.choices)
```

```
[Choice(finish_reason='length', index=0, logprobs=None, message=ChatCompletionMessage(content='The OpenAI API is a cloud-based service provided by OpenAI that allows developers to integrate advanced AI models into their applications.', refusal=None, role='assistant', function_call=None, tool_calls=None))]
```

- List with one element

Interpreting the response

```
print(response.choices[0])
```

```
Choice(finish_reason='length', index=0, logprobs=None,  
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role='assistant', function_call=None, tool_calls=None))
```

Interpreting the response

```
print(response.choices[0].message)
```

```
ChatCompletionMessage(content='The OpenAI API is a cloud-based service provided by OpenAI that allows developers to integrate advanced AI models into their applications.', refusal=None, role='assistant', function_call=None, tool_calls=None)
```

Interpreting the response

```
print(response.choices[0].message.content)
```

The OpenAI API is a cloud-based service provided by OpenAI that allows developers to integrate advanced AI models into their applications.

- Response as a string

Interpreting the response



LET'S PRACTICE!

